

Beating the odds in the innovation arena: The role of market and technology signals classification and noise

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Abstract

Innovation is a risky business. Notwithstanding the attention in research on the determinants of innovation success, the estimates of the extremely high failure rates (ranging from 40% to 90%) have not come down. As a result, new approaches have to be found to address the issues that prevent organizations from reaching innovation success over a longer period of time. In this study, Signal Detection Theory (SDT) is introduced to show that it may be less important to improve innovation practices in companies, than it is to change the nature of the projects that enter the corporate innovation funnel. In our empirical study of 44 innovation projects in the pharmaceutical and electronics industry, we show that pursuing more noisy signals (uncertain technological and market opportunities) is currently beneficial for innovation success. In general, companies allocate too few resources to noisy innovation opportunities. Interestingly, we find a positive, and not an inverted U-shaped, relationship between noise/signal ratios and innovation success. This indicates that there are almost no companies that take too much risk in the current competitive environment and that many can gain from increasing their noise tolerance. The implications from this study are quite counter-intuitive in a sense that 'risk' is not the cause of the innovation problems, but may be a solution.

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1. Introduction

Innovations are the life-blood of most firms (Shane, 2000; Wind & Mahajan, 1988). Yet, while it appears that many companies only develop some successful innovations by accident, a few other firms seem to have a consistent supply of rewarding innovations (Crawford & Di Benedetto, 2003). For instance, the Minnesota Mining and Manufacturing Company (3M) is widely recognized as a truly successful and innovative company. Today, 3M makes 66,000 products as diverse as fire hose linings, surgical gowns, bingo supplies and Scotch tape (3M Annual Report 2003). The company consistently ranks in the top 10 of Fortune's list of most admired companies and makes more than a quarter of its sales

from products less than 4 years old. In the biotechnology industry, companies like Amgen have become industry leaders with a consistent flow of new products such as *Epogen*, while other companies in the industry are taken over by others because they have no new products to replace their older ones, which are running out of patent protection.

The potential causes for innovation performance differences have been widely established in earlier research, such as: variations in the quality of market analysis, differences in product design capabilities, timing, and marketing efforts (Griffin & Hauser, 1996; Henard & Szymanski, 2001; Ottum & Moore, 1997). From a strategic marketing perspective, many companies seem to have trouble accumulating and exploiting scarce innovation resources. Scientists adhering to the resource-based views argue that companies have to *acquire* and *leverage* scarce and difficult to imitate resources and capabilities to establish a sustainable competitive advantage (Barney, 1991; Day, 1994; McEvily, Eisenhardt, & Prescott, 2004). Interestingly, while much research is available and knowledge grows

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on why some firms are more innovative than others, the difficulties surrounding innovation hardly seem to disappear (Cooper, 1993; Crawford & Di Benedetto, 2003; Montoya-Weiss & Calantone, 1994).

One of the key reasons for failure seems to be inherently linked to the uncertainties surrounding any new product and service. That is, before technological and market changes lead to new products and services, organizations have to identify and prioritize opportunities and allocate resources to develop them further. Since opportunities are surrounded with different degrees of uncertainty, opportunity identification and classification are far from trivial (Shane, 2000). Interestingly, the identification of an opportunity may not come from active search by an individual or organization. It may come from the recognition of the value of new information that is received through open channels with the environment (Kirzner, 1997). Since the recognition is based on prior knowledge and experiences of the people in the organization, only a subset of organizations will possess a background that triggers the opportunity identification (Shane, 2000).

In the present study, we introduce and test a relatively unknown theory from physics and psychology named Signal Detection Theory (SDT) that can complement current insights on innovation performance. SDT provides a theoretical approach to the identification and selection of signals that are pursued for further action by individuals or organizations. SDT is now emerging as a modeling tool in marketing and other business areas, such as advertisement recognition testing (Cradit, Tashchian, & Hofacker, 1994), sales effectiveness (Knowles, Grove, & Keck, 1994), auditing (Karim & Siegel, 1998), and competition (Lampel & Shapira, 2001). This theory seems to be very valuable in relation to existing resource-based research on openness and absorptive capacity as it provides a clue as to why companies may, or may not, pursue certain market and technology signals (Cohen & Levinthal, 1990).

SDT is a model on how humans detect (filter) signals in a background of interference or noise. Noise blurs the signal and is therefore a burden that needs to be reduced. However, from a marketing and innovation performance perspective, which is the focus of this study, noise may not be a burden at all since it may distract competitors from detecting the same signal. This may lead to an early product introduction and probably a better performance in the market. However, as will be discussed later, there are also other stakeholders that often prevent organizations from pursuing noisy signals. All these resulting forces together result in the fact that organizational filters range from ‘too loose’ (resulting in too many dead ends and costs) to ‘too rigid’ (resulting in too few unique competitive breakthroughs).

The next section of this paper discusses the concept and aspects of learning and SDT. We test our framework with data from two industries, namely consumer electronics and pharmaceuticals ($n=44$). After that, we analyze our results and present our findings. Finally, the managerial implications, and suggestions for further research are presented.

2. Theory on learning and SDT

The process of knowledge acquisition and leveraging is often labelled “organizational learning”. For example, Slater and Narver (1995) define the process of learning as a three-stage process that includes information acquisition, information dissemination, and shared interpretation. Others conceptualize the organizational learning process as the acquisition, dissemination, and utilization of information (Argote, 1999). While in these definitions *information* is in the core, the purpose of learning is related to *knowledge* improvement or *knowledge* creation. Nonaka and Takeuchi (1995) state: “learning is defined as the activities through which group members acquire, share, and combine knowledge into a collective product through experience with another”. As a facilitator, a learning orientation of the organization consisting of a high commitment to learning, shared vision, open mindedness, and adequate knowledge sharing, is needed (Calantone, Cavusgil & Zhao, 2002).

The theory of knowledge creation is anchored to the assumption that human knowledge is created and expanded through the social interaction between tacit and explicit knowledge. Prencipe and Tell (2001) argue that: “the quintessential knowledge creation process takes place when tacit knowledge is converted into explicit knowledge. In other words, our hunches, perceptions, mental models, beliefs, and experiences are converted to something that can be communicated and transmitted in formal and systematic language between people”. Given the uncertainties and limited data in innovation, this process plays an important role in innovation in general.

From a broader resource-based perspective, the term acquisition and leveraging refer to distinctive types of activities that can engender competitive advantage from technological and market-based competencies. Acquisition is the process by which organizations absorb, develop, and renew competencies. Leveraging encompasses the transfer of knowledge to different organizational units and the exploitation, integration, and recombination of extant competencies (McEvily et al., 2004). Learning is therefore a key organizational phenomena that is often conceptualized as a sequential or linked process within and between functional groups. As a result, understanding how *groups* learn, provides the micro foundation for understanding organizational learning (Hatch, 1997). A group is made up of individuals who share some identity and whose tasks are interdependent. As a result, they have to show some level of cooperative behaviour to be successful. Many different groups exist in organizations and they can be large or small, temporarily or for a very long time. What is especially important in innovation are functional and cross-functional groups (Griffin & Hauser, 1996). Functional groups provide ‘depth’ in a certain functional area such as R&D or Marketing (Allen, 1977). The use of cross-functional groups facilitates integration and time to market for developing new products (Crawford & Di Benedetto, 2003, p. 303). As a result, innovation performance is to a considerable extent the result of a long process of information or knowledge acquisition and sharing within and between functional groups. So, successful innovation over a longer period of time has to be facilitated with a learning process that scores high on a ranges of key metrics or strategic

success factors such as (see Brockhoff, 2003): (1) information acquisition, (2) group learning, and (3) sharing (also with other companies and customers).

Before we will identify some key learning/innovation metrics and develop hypotheses, Signal Detection Theory is introduced.

2.1. SDT

Signal Detection Theory provides a theoretical approach to the identification and selection of signals. SDT is a model on how humans detect signals in a background of interference or noise. The human observer considers a particular event and decides whether it is a signal or noise. SDT assumes that there is an overlap between the distributions of signal and noise, so that a particular observation may have come from either distribution. The observer then has to decide whether to accept or reject the event as a signal. There are four possibilities in the decision matrix of the observer (Cradit et al., 1994):

1. the observer notices a noise when it is a signal (miss)
2. the observer notices a signal when it is a signal (a hit)
3. the observer notices a noise when it is a noise (correct identification)
4. the observer notices a signal when it is a noise (false alarm).

The observer should make a decision concerning the event and classify it either as a signal or a noise. There are two main components in the decision-making process: information acquisition and criterion. The purpose of acquiring more information is to increase the likelihood of getting either a hit or a correct identification. The second component is subjective; it is based upon the own judgement of the observer. This judgement might be based on existing knowledge capabilities, mental models, and premises. From a corporate perspective, certain signals can only be detected by adjusting the noise tolerance.

There is a major difference between business and physics: in physics information has the role of eliminating noise as much as possible. By eliminating noise in physics, the level of risk in decision-making is decreased. In business, noise is almost a necessary element as without noise the signal can be detected by any competitor, which makes it a less attractive new product opportunity. The ability of being able to detect and act upon (weak) signals in an environment with large amounts of noise, before others do, can lead to a competitive advantage because the market will be less crowded and/or positions can be better defended (Day, 1994). This argument also relates to the competitive forces approach of Porter (1980) and it puts an emphasis on the intensity of competition in the industry or market segment that determines profitability.

2.2. Hypotheses

As mentioned before, information and knowledge leveraging are considered two key components of the learning process and ultimately innovation success. Relevant information flows can come from a variety of sources, including internal databases,

which contain to a certain extent the firm's organizational memory, and external sources such as journals, reports, and from interactions with other firms (Mascitelli, 2000). In an innovation context, the detection and generation of relevant market and technology information is linked to resources such as absorptive capacity or open relationships with the environment (Cohen & Levinthal, 1990). Organizational openness relates to the mere availability of relevant information through a range of channels and this information is later transferred through face-to-face interactions and through dialogue to other employees of groups. Since the core groups in innovation are *functional* (specialized in for example basic science) or *cross-functional* (such as a cross functional team), the following hypotheses are formulated as a starting point:

H1. There is a positive relationship between openness of the corporate innovation system and innovation success.

H2. There is a positive relationship between knowledge leveraging *within* and *between* functional departments and innovation success.

Signal detection theory argues that the interpretation of all (new) information depends on the sensitivity of the observer. As a result, the availability of information alone and capable downstream leveraging capabilities are not sufficient; the noise filter determines whether a piece of information is detected at all and classified as an opportunity. The level of success in filtering signals in a background of interference of noise is believed to be dependent on what we will call the filter sensitivity (see Fig. 2).

Filter sensitivity relates to the firm's degree of sensitivity with respect to responses to market and technology signals. So, it indicates whether a firm has a low or high threshold level to respond. This dimension relates to the firm's willingness to take risk. In cases of high noise levels (upper part of Fig. 2), where signals are really blurred, there can be two types of outcomes: (1) a competitive miss (no action was taken) or (2) a competitive hit (action was taken). In cases of low noise (lower part), a competitive hit is not likely to be obtained because other companies see the same opportunity, resulting in me-too competition. If there is no uncertainty and the company decides not to take action, there was likely no sound business case to pursue.

Fig. 2 shows the application of SDT in competitive innovation. If one compares it with Fig. 1, there are two

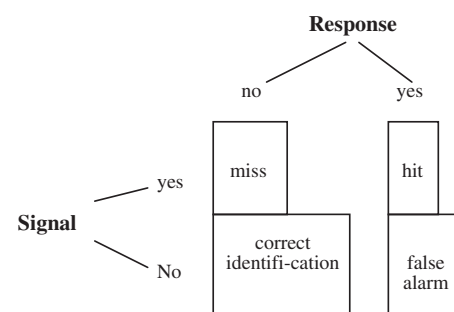


Fig. 1. The Signal Detection Theory (SDT) model.

major differences. First, there is no signal and noise distinction because there is always an opportunity (signal) behind innovation work and as a result there is only more or less noise (vertical axe). Second, there is a special cell called “competitive hit”, which relates to the fact that competitors may not pursue the signal as well, likely to lead to above average returns in the market (Porter, 1980). It is important to note that more noise is not similar to a more innovative product. Innovativeness is related to the technologies or products that are developed to address the opportunity. Innovative or less innovative products may be developed for all quadrants.

An interesting question is what happens if a company adopts more noisy signals? Fig. 2 shows that these types of companies position themselves in the competitive hit arena (lower figure). A noise neutral innovator will be equally active in all four quadrants (upper figure). A noise tolerant firm will be biased towards the upper right cell (competitive hits).

There are many forces that determine the level of signal/noise sensitivity an organization wants, or can, achieve in its innovation efforts. Stakeholder theory (Freeman, 1984) suggests that organizations have to manage many relationships in an unpredictable environment and many stakeholders may not

like the focus on more noisy signals. Investors and credit rating agencies, for example, may not prefer these initiatives. Others, such as innovative partners and entrepreneurial employees, may welcome a higher noise tolerance as illustrated by the following case description:

Box 1

Epogen of the biotechnology company Amgen

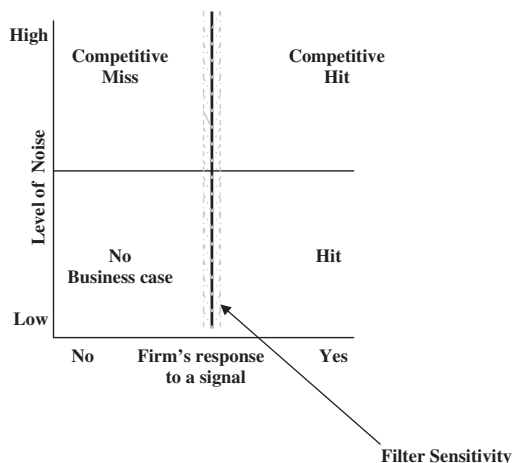
Since its foundation in 1980, the company Amgen has grown from a handful of employees at its Thousand Oaks headquarters in California, to a company employing more than 10,000 people spanning the globe in 2002. Amgen discovers, develops, manufactures and markets human therapeutics based on advanced cellular and molecular biology. Its annual revenue is around \$ 6 billion in 2003.

The success of Amgen started with Epogen. Epogen is a recombinant erythropoietin product. Erythropoietin is a protein hormone that originates from the kidney and stimulates red blood cell formation. Patients with impaired kidney function do not produce sufficient quantities of erythropoietin and, as a result, develop chronic anemia. Without adequate amounts of erythropoietin, there are too few red blood cells to deliver oxygen throughout the body.

Amgen was the only company to pursue and develop a product based on the hormone. It seems that Amgen's filter sensitivity allowed it to detect and pursue the signal and to isolate and synthesize the human erythropoietin gene, which was extremely difficult with no guarantee for success. Other firms also thought there was no big market. However, the leading scientists devised a clever way of isolating the human gene based on a new method of screening human genomic libraries. After two years of long days and nights in the laboratory they succeeded in isolating the human gene and this placed Amgen in the leadership position worldwide to develop a therapeutic agent. After that, the market exploded as tests showed beneficial results for all types of patients (and illegally for athletes).

Sources: www.Amgen.com, Amgen press releases.

Noise Neutral Innovator



Noise Tolerant Innovator

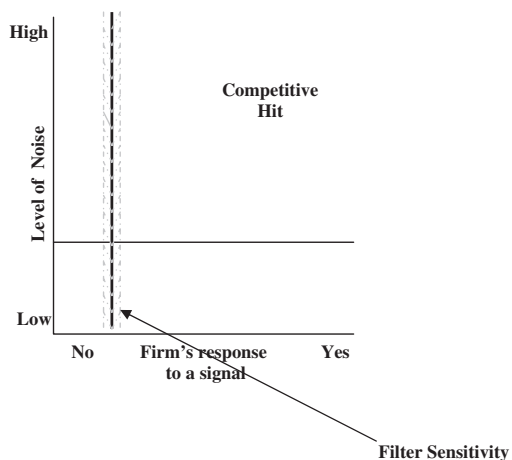


Fig. 2. SDT models for two types of organizations.

We recognize that at some point, setting the right level of noise tolerance becomes a delicate balancing act with respect to all stakeholders.¹ For example, if filters are too loose, (too) many signals will be pursued, leading to many dead ends and wasted resources that could paralyze a firm's innovation system

¹ There are many strategies a firm can follow with respect to secretly hiding initiatives and signaling that it is, or is not, pursuing certain signals. This is part of how companies try to protect advantages or try to create a lead-time advantage. However, in this study we focus on what is actually pursued in the NPD funnel.

Table 1
Overview of projects in the sample (as perceived by the respondent)

	Breakthrough	Incremental	Total
Pharmaceuticals	14	10	24
Electronics	3	17	20
Total	17 (39%)	27 (61%)	44 (100%)

and performance. If filters are too tight, insufficient competitive hits will occur, leading also to suboptimal performance as well. However given the low levels of noise tolerance of important stakeholder such as banks and shareholders, we expect that in the current environment, accepting more noise is better for performance. So we expect that:

H3. In the current business environment, there is a positive relationship between noise/signal (tolerance) and innovation success.

By conducting a comprehensive test of the hypotheses, we hope to be able to show that SDT's key noise concept can offer additional explanatory power in explaining innovation success. As such, it can establish SDT in an innovation context.

3. Methodology

3.1. The research population and sampling

The empirical part of this study was conducted in two industries: pharmaceuticals and electronics. We opted for these industries for several reasons. First, innovation plays an important role in these industries and relatively many companies are involved in innovation activities. Second, these companies pursue projects with high and low levels of noise around market and technology signals. Finally, the two industries cover a *process* industry and an *assembly/manufacturing* industry. It seems interesting to explore whether there are similar mechanisms behind success. The final data set consists of 24 biotechnology projects and 20 electronics projects in the Netherlands and Belgium. The projects range from recombinant human insulin and a HIV test to a large Xenon car light project. The average project size is \$25 million Euros with the most expensive one being an estimated \$200 million Euros. The sample was developed by contacting firms in this industry randomly from a large list of companies mentioned in trade journals and identifying project

managers that were willing to participate in the interviews. The final response rate of companies in the sample was 80%.

3.2. Questionnaire

Data were collected by means of semi-structured interviews of about 1–1.5 h at the company site. With respect to the open parts of the interview, notes were taken and they were later send to the informant to provide feedback. The structured parts were developed using multi-item scales for the key variables. The unit of analysis for the questionnaire was the new product development project and the informant was asked to select the *most recent project* he/she worked for. We decided to ask for the most recent instead of successful/unsuccessful to get a more random draw of projects and to reduce memory problems.

3.3. Measures

The key measures in this study are: (1) a set of variables related to “openness of the innovation system”, (2) a set of variables related to “knowledge leveraging”, (3) “noise/signal ratio”, and (4) “innovation success”. We used existing scales as much as possible and a complete list of the items and their source is presented in Appendix 1. We decided not to use measures for the strength of the signal and noise separately because there was a strong negative correlation between signal and noise in the data. As a result, a comprehensive noise/signal measure was used with higher scores indicating more noise relative to the signal. Scale analysis was used to make sure that the measures are uni-dimensional with sufficient reliability. The variable “Resources” was included as a control variable. The Cronbach α values of the multi-item scales are acceptable (see Appendix 1).

3.4. The projects in the sample

The response set contains a broad variation of projects, including breakthrough and incremental innovations (as perceived by the respondent).

Table 1 provides an overview of the level of innovativeness of the projects that resulted from our sampling approach. The projects of the sample are relatively important to the companies concerned: 77% of all the projects are perceived by the respondents to belong to the (future) core of the company.

Table 2
Descriptive statistics and inter variable correlations

	1	2	3	4	5	6	7	8	9
1 Internal knowledge acquisition	1								
2 External knowledge acquisition	.022	1							
3 Functional learning	.116	.238	1						
4 Cross-functional learning	.105	.290	.035	1					
5 Learning from customers	.242	.112	.403 *	.105	1				
6 Learning from other firms	-.162	.112	.094	.101	.298 **	1			
7 Noise/Signal	.101	-.086	-.307 **	.162	.003	.158	1		
8 Resources	-.003	-.171	.217	-.031	-.186	-.019	-.405 *	1	
9 Innovation Success	.273 **	-.160	-.119	-.013	.330 **	.282	.449 *	-.271	1

* Correlation is significant at the 0.01 level (two-tailed).

** Correlation is significant at the 0.05 level (two-tailed).

4. Results

Before we test our hypotheses, the inter-variable correlation matrix is presented in Table 2.

The correlations matrix reveals that there is a high positive correlation between functional learning and learning from customers. This is in line with the general trend to use customer feedback, customer visits, and prototyping in innovation. The correlation matrix also shows some negative relationships between variables. A negative correlation is revealed between noise/signal and functional learning (intra departmental). This may indicate that specialized groups are less interested to develop new knowledge based on weak signals surrounded by considerable amounts of noise. For marketing, it is often the case that the absence of upfront market potential might negatively affect the eagerness of marketers to ‘go for it’. A negative relationship is also found between resources and noise ($r = -.405$, $p < .01$). It is important to note that resources were measured at the project level and not at the firm level. A negative correlation between resources and noise could indicate that projects with higher noise levels (competitive hit area of the SDT model for innovation) are rewarded with a relatively lower level of resources. Possibly management is having some reservations (related to risk and uncertainty) when awarding project ideas with resources that are based upon relatively weak signals. As such, the low levels of allocated resources could resemble ‘probing’ before making definite choices, but it may also be a sign that the noise tolerance is relatively low on average. Finally, the correlation matrix shows that there is a *positive* correlation between ‘innovation success’ with ‘learning from customers’ and ‘noise’.

As a second step, a multiple-regression model is estimated. Based on the correlations between the independent variables ($r < .5$) multicollinearity is not a major problem. To test for a non-linear effect of noise, we included the main effect for noise and (noise $\times$ noise). To reduce multicollinearity further, the variables noise and noise $\times$ noise were mean centred (Jaccard, Turrisi & Wan, 1990).

4.1. Factors that affect innovation success

Four independent variables have a significant effect on innovation success: internal knowledge acquisition, learning from other firms, cross-functional learning, and noise/signal.

4.1.1. Openness

Internal knowledge acquisition is significant ($\beta = .396$, $p < .05$). It shows that proper and active consultation of up-to-date, reliable and accessible internal knowledge – that holders are willing to share – has a positive influence on innovation success. Earlier research has also stressed the importance of organizational memory that, among others, is embedded in the organization’s knowledge base (Argote, 1999). External knowledge acquisition is not significant (the negative sign may indicate that there are counter-productive forces in terms of protecting resources if firms depend too much on outside knowledge). *Learning from other firms* is significant and

positive with respect to innovation performance ($\beta = .357$, $p < .05$). It highlights the importance of outside learning that results from interactions with other firms and openness. It reveals that learning from other firms, such as co-developing firms and also universities or research centres, highly contribute to success. H1 is generally supported.

4.1.2. Leveraging knowledge: functional learning and cross-functional learning

The table shows a positive but not significant β for *functional learning*. So, although the direction is in line with our expectations, it does not significantly explain variance in innovation performance.

Cross-functional learning is significant, but it has a *negative* coefficient ($\beta = -.369$, $p < .1$). Possibly interface management is over-emphasized in the projects in the sample. In general, cross-functional interfaces are considered critical in the new product development process. It is not unlikely that in their efforts to manage the critical interfaces between for instance Marketing and R&D, the target has been overshot. To put it differently, possibly there is ‘too much cross-functional talking and cooperation’ (for instance between departments or in multi-functional project teams). Souder (1988) for example found that there were organizations where marketing and R&D were too good friends. In addition, Leenders and Wierenga (2002) found that there are some negative performance side-effects of certain (formal) organizational integration mechanisms. H2 is not supported.

4.1.3. Noise/signal

Noise/signal is significant and the coefficient is in the expected direction ($\beta = .326$, $p < .1$). The regression analysis confirms that noise contributes to innovation success. It may indicate that noise hides weak signals, which are harder to detect, but if pursued lead to more successful new products because other firms do not follow the same track, resulting in less (me-too) competition and higher market performance. So,

Table 3
Multiple regression analysis including a curvilinear noise effect

Independent variable:	β	t	Significance
<i>Openness</i>			
Internal knowledge acquisition	.396	2.389	.025
External knowledge acquisition	-.074	-.460	.650
Learning from customers	.151	.873	.391
Learning from other firms	.357	2.094	.047
<i>Leveraging</i>			
Functional learning	.107	.572	.572
Cross-functional learning	-.369	-1.898	.069
<i>Noise tolerance</i>			
Noise/signal (mean centred)	.326	1.798	.084
Noise $\times$ noise (mean centred)	.045	.220	.827
<i>Control variable</i>			
Resources	-.088	-.421	.678

Dependent variable: innovation success.
 $R^2 = .28$.

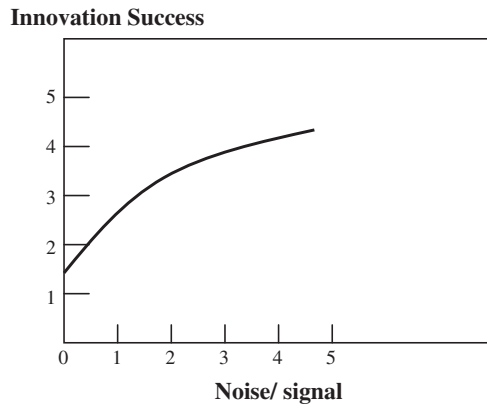


Fig. 3. Relationship between noise/signal and innovation success.

the drawbacks of a more lenient noise classification system (e.g., information overload, costs of trying many avenues) do not seem to outweigh the potential benefits.

4.1.3.1. Exploring possible non-linear relationships between noise and innovation success. It may be the case that noise/signal has a non-linear effect on performance. In other words, there may be a curvilinear effect or even an optimum. Table 3, however, does not show a significant coefficient for the noise $\times$ noise variable after controlling for the main effects. The increase in R^2 due to the interaction is also not significant. So, there does not seem to be an optimal level of noise with respect to performance (inverted U-shape): Increases of noise are accompanied by increases in performance and there is no turning point where noise starts to influence innovation success negatively (see Fig. 3). H3 is supported.

The curve estimation reveals that the curve has an almost linear progression. This is a quite remarkable result because there is quite some variation in the level of noise in the sample. So, it shows no real limit to the benefits of pursuing noisy signals in the current industry environment. On the contrary, the more noise the better. The curve shows an almost linear relationship between noise and new product performance. Of course, theoretically it is still possible that the figure is different for more extreme levels of noise or no noise at all. However, our study shows that companies may increase innovation performance by selecting signals that are less clear and are surrounded with more noise. From a stakeholder perspective, the results show that noisy signals may not only be pursued from a competitor perspective, but also from a financial or shareholder perspective as the benefits outweigh the risk.

Different relationships between noise/signal and performance may exist beyond the noise level included in this study. Maybe the most likely relationship is that an optimal noise level occurs somewhere (after that, it really becomes too risky and involves too much costs). Another possibility may be that the curve would show a steep increase beyond a certain level of noise. This could occur in such cases where no single competitor pursues the signal. This was for example the case in the pharmaceutical industry when the company AstraZeneca developed the drug *Losec*. The ethical drug completely blocks the acid production in the stomach to prevent stomach problems. Because all companies

thought this was too rigid to work (noise), it was not pursued. However, it created one of the most financially successful products ever developed in the pharmaceutical industry. Finally, the curve could show a drop-off after our signal/noise levels. This could occur in case that the risks and costs are really too high for larger levels of noise relative to the signal.

Finally, we conducted several sensitivity analyses. First, we added innovativeness to the regression as an independent variable. Innovativeness was positively related to innovation success ($\beta = .28, p < .10$) but did not substantially affect the other coefficients. The correlation between noise and innovativeness is $r = .24, p < .1$ supporting the notion that noise and innovativeness are different concepts. To test whether there are substantial differences between the two industries, we ran separate regressions for the two industries. However, we found similar patterns in size of the coefficients and their signs. In addition, independent samples *t*-tests were done to see whether the industries differ on the main variables in this study. Overall, we only found significant differences in the average scores for external knowledge acquisition (mean Electronic Industry = 3.15, mean Biotech = 3.76) and for resources (mean Electronic Industry = 4.05, mean Biotech = 3.29). This indicates that the pattern in the data is quite general and robust.

5. Conclusions and discussion

The starting point for organizational learning in new product innovation is often an opportunity or idea with respect to new technologies, emerging needs in the market, or both. In this research we adopted a relatively new approach to explain why some firms are more successful with their innovation efforts than others: Signal Detection Theory (SDT).

Bringing successful innovations to the market involves a long process of information processing and learning. SDT shows that the chances for success are related to the level of noise that firms find acceptable (related to market and technology risk/uncertainty). Interestingly, noise tolerance has two sides. If a lot of uncertainty is accepted (low signal to noise opportunities are pursued), a lot of investments have to be made and many dead ends have to be taken for granted. This may not be in line with the interests of important stakeholders such as banks or credit rating organizations. If only low levels of uncertainty are accepted, many missed (not recognized) signals will also result in low innovation performance. In addition, certain stakeholders such as entrepreneurial employees or network partners will not prefer this approach either. The key question that results from this is where the balance currently lies: Do companies accept too much or too little noise?

From our empirical research in the pharmaceutical and electronics industry, we learned that companies may currently be a little too conservative with the levels of noise they find acceptable – or, to put it differently, that the corporate filters are too rigid and that they are missing many interesting opportunities. A closer look at the relationship between the level of noise around the signal and innovation success even showed that there is currently a linear positive relationship between noise and new product success.

Noise is a combination of all kinds of surplus and non-relevant technical and market information that distracts an observer from identifying a signal that could be pursued – if detected – and could lead to a successful new product. Noise is making it more difficult to detect a signal, but, maybe a bit paradoxical, this very nature of noise affects new product performance. Noise contributes to new product success because noise ‘hides’ signals and therefore decreases the likelihood that competitors pursue the same learning track. Thus, if competitors do not pursue learning it can create a competitive advantage for others (Porter, 1980). In environments where successful new products are, to a large extent, based upon achieving and owning first-to-market intellectual property, noise is even a pre-condition for new product success. This is quite counter-intuitive in a sense that innovation risks are not the cause of the innovation problems and low success rate, but a competitive opportunity.

Besides detection of the signal, translating the signal into appropriate action is important too. Here we believe entrepreneurship may play an important role. Weak signals often result in weak business cases, that, if rewarded at all, often lack sufficient resources for action because management might tend to prioritize towards stronger, and less risky business cases. The entrepreneur, driven by personal belief and commitment, also plays a major role in this arena through the ability of coalition forming. The importance of the individual, or a small group of people, in detecting and pursuing the weak signal is striking in the *Epogen* case presented earlier. Research on entrepreneurs and opportunity identification strongly supports this notion. Shane (2000) argues that more research is needed on why some entrepreneurs manage to exploit some technologies and markets. He argues for example that “entrepreneurs can and will discover those opportunities without searching for them”. This may have a direct link to SDT. Our interviews strongly showed that the entrepreneurs were noise tolerant and able to combine information from the market and from technology, which led to the recognition of the weak signal out of noise.

Most of the time, entrepreneurs followed their personal belief in pursuing a signal based on his or her prior knowledge (Shane, 2000). Personal belief and commitment is an important driving force, as an entrepreneur has to overcome internal skepticism (existing filters) that results from noise (‘this is not possible’, ‘there is no market for this’, and so on) and many times has to fight for resources. Obtaining resources for the new idea based upon weak signals can be very difficult as weak signals mostly also lead to weak business cases. Weak business cases lead to difficulties in getting financial support to start up a new product project. Entrepreneurs however do not let themselves hinder by this, they are willing to take risks to find the resources they need. These entrepreneurs, either individuals or small groups of people, have a specific capability to get things done, such entrepreneurs can be found in any organization. These entrepreneurs are capable of coalition forming and finding other individuals in pursuing and accomplishing their goals. So, more research is needed to understand this process (see also Daniel, Hempel, and Srinivasan (2003) for a broader discussion on project selection).

Overall, the findings of this research contribute to improving new product innovation success by increasing the awareness that there is a rewarding side to the tolerance for – and the ability to detect and pursue – noisy signals in the innovation arena.

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Appendix A. Overview of measurement scales

Internal knowledge acquisition

Items	Source
1. Information sources inside the firm were easy to access	Moorman and Miner (1997)
2. Information inside the firm was reliable	Moorman and Miner (1997)
3. Information inside the firm was updated	Moorman and Miner (1997)
4. Holders of information were willing to share and cooperate	Moorman and Miner (1997)

Cronbach α reliability of 0.7469.

External knowledge acquisition

Items	Source
5. The knowledge potential of external information sources (literature, universities, internet) was known and exploited	Kessler, Bierly, and Gopalakrishnan (2000)
6. External information sources were easy to access	Kessler et al. (2000)
7. External firms (customers, suppliers, others) were an important source of information	Kessler et al. (2000)
8. External information was reliable	Kessler et al. (2000)

Cronbach α reliability of 0.7162.

Learning from customers

Items	Source
9. Feed-back and learning from customers were actively pursued	Lynn, Morone, and Paulson (1996)
10. Some form of prototyping and/or test projects, including iterative feed-back loops with customers, were used in the project	Lynn et al. (1996)
11. Customers were/are open and willing to share	Lynn et al. (1996)
12. Customers provided both explicit and tacit knowledge	Mascitelli (2000)
13. Tacit knowledge from customers was important	Mascitelli (2000)

Cronbach α reliability of 0.7597.

Learning from other firms

Items	Source
14. There was involvement of other firm(s) in the development of the product	Tsai (2001)
15. Other firm(s) were developing a complementary product	
16. The level of openness of the other firm(s) was appropriate	Tsai (2001)
17. The level of trust between the two firms was appropriate	Tsai (2001)

Cronbach α reliability of 0.6806.

Functional learning

Items	Source
18. With this project a lot was learned within Research	Lynn, Skov, and Abel (1999)
19. With this project a lot was learned within Development	Lynn et al. (1999)
20. With this project a lot was learned within Marketing	Lynn et al. (1999)
21. Problem solving and trial-and-error (based upon today's technology) within the department were an important source of knowledge creation	Lynn et al. (1996)
22. Experiments within the department were an important source of (new) knowledge creation	Lynn et al. (1996)
23. Tacit knowledge within the department was important for learning	Mascitelli (2000)

Cronbach α reliability of 0.6814.

Crossfunctional learning

Items	Source
24. Knowledge transferred easily from Research to Development	Ottum and Moore (1997)
25. Knowledge transferred easily from Development to Marketing	Ottum and Moore (1997)
26. Knowledge transferred easily from Research to Marketing	Ottum and Moore (1997)
27. It was possible to by-pass certain departments to gather new knowledge	Ottum and Moore (1997)

Cronbach α reliability of 0.7714.

Noise/Signal (higher scores mean more noise relative to the signal)

Items	Source
28. The idea for the new product came from a weak technological signal	Shane and Venkataraman (2000)
29. The idea for the new product came from a weak market signal	Shane and Venkataraman (2000)
30. Competitors clearly missed the signal that led to this innovation	Shane and Venkataraman (2000)

Cronbach α reliability of 0.6519.

Noise/Signal Innovation success

Items	Source
37. The project was a success with regard to costs	Cooper (1993)
38. The project was a success with regard to time	Cooper (1993)
39. The project was a success with regard to Return On Investment	Cooper (1993)
40. The project was a success with regard to competitiveness	Cooper (1993)

Cronbach α reliability of 0.6043.

Resources (single item)

Item	Source
41. Availability of resources for the project was no issue	Argote (1999)

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